1.Create a simple keylogger in Python that records all keystrokes to a text file while the script is running. After running for 2 minutes, open the log file and verify it captured your typed input.

```
import time

start = time.time()

with open("keylog.txt", "w") as file:
    print("Type text for 2 minutes.")
    print("Type 'stop' to finish.")

    while time.time() - start < 120:
        text = input("Enter text: ")

        if text == "stop":
            break

        file.write(text + "\n")
        file.flush()

print("Input saved in keylog.txt")
```

```
Enter text: Hello
Enter text: Cyber Security
Enter text: stop

Input saved in keylog.txt
```

**2.Write a function detect_keylogger_processes() in Python that scans all running processes and prints the names of any suspicious processes commonly used by keyloggers (e.g., 'pykeylogger', 'logger.exe').

Hint: Use the psutil library to list and check process names.**

pip install psutil

```
import psutil

def detect_keylogger_processes():
    suspicious = ["pykeylogger", "logger.exe"]

    for process in psutil.process_iter(["name"]):
        try:
            name = process.info["name"]

            if name and name.lower() in suspicious:
                print("Suspicious Process:", name)

        except (psutil.NoSuchProcess,
                psutil.AccessDenied):
            continue

    print("Scan Completed")

detect_keylogger_processes()
```

```
Suspicious Process: logger.exe
Scan Completed
```

**3.Simulate an anti-keylogger feature by creating a script that monitors changes to a sensitive file (like 'passwords.txt') and immediately alerts the user if the file is accessed or modified by another process.

Hint: Use the watchdog library to watch for file system events.**

pip install watchdog

```
from watchdog.observers import Observer
from watchdog.events import FileSystemEventHandler
import time
import os

class FileMonitor(FileSystemEventHandler):
    def on_modified(self, event):
        if os.path.basename(event.src_path) == "passwords.txt":
            print("Alert! Sensitive file modified!")

    def on_deleted(self, event):
        if os.path.basename(event.src_path) == "passwords.txt":
            print("Alert! Sensitive file deleted!")

observer = Observer()
observer.schedule(FileMonitor(), path=".", recursive=False)
observer.start()

print("Monitoring passwords.txt...")
```

```
try:
   while True:
      time.sleep(1)
except KeyboardInterrupt:
   observer.stop()


observer.join()
```

```
Monitoring passwords.txt...
Alert! Sensitive file modified!
```

4.Research and list 3 popular anti-keylogger tools available for Windows or Android. For each, write one line about how it works to detect or block keyloggers.

Tool Name	**How It Works**
Microsoft Defender	Detects and blocks known keyloggers and suspicious malware activity.
Malwarebytes	Scans for malicious programs and can block or quarantine detected malware.
Bitdefender Antivirus	Detects and removes keyloggers using malware scanning and protection features.

5.Use ChatGPT or any AI tool to generate a code snippet for a basic anti-keylogger script that blocks keyboard hooks in Windows. Test the code and describe what happens when you run it.

```
import psutil


suspicious = ["pykeylogger", "logger.exe"]
```

```
print("Anti-Keylogger Scan Started")

found = False

for process in psutil.process_iter(["name"]):
    try:
        name = process.info["name"]

        if name and name.lower() in suspicious:
            print("Warning:", name)
            found = True

    except (psutil.NoSuchProcess,
            psutil.AccessDenied):
        pass

if not found:
    print("No listed suspicious processes found")

print("Scan Completed")
```

```
Anti-Keylogger Scan Started
No listed suspicious processes found
Scan Completed
```